

Évolution mémoire : Tableaux et débordements (64 bits)

Étape 0: int a[6]; /* a = 0x10 */

64-bit example
(pointers are 64-bits wide)

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10							
a[2]	0x18							
a[4]	0x20							
0x28								
0x30								
0x38								
0x40								
0x48								
0x50								

Étape 1: a[0] = 0x015F; a[5] = a[0];

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10	5F	01	00	00			
a[2]	0x18							
a[4]	0x20				5F	01	00	00
0x28								
0x30								
0x38								
0x40								
0x48								
0x50								

Étape 2: a[6] = 0xBAD; /* DÉBORDEMENT ! */

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10	5F	01	00	00			
a[2]	0x18							
a[4]	0x20				5F	01	00	00
0x28	AD	0B	00	00				
0x30								
0x38								
0x40								
0x48								
0x50								

⚠ Buffer
Overflow!

Étape 3: a[-1] = 0xBAD; /* Pas d'erreur !!! */

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10	5F	01	00	00			
a[2]	0x18							
a[4]	0x20				5F	01	00	00
0x28	AD	0B	00	00				
0x30								
0x38								
0x40								
0x48								
0x50								

⚠ Écrit AVANT
le tableau!

Étape 4: int *p = a; *p = 0xA;

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10	0A	00	00	00			
a[2]	0x18							
a[4]	0x20				5F	01	00	00
0x28	AD	0B	00	00				
0x30								
0x38								
p	0x40	10	00	00	00	00	00	00
0x48								
0x50								

Étape 5: p[1]=0xB; p=p+2; *p=a[1]+1;

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x00	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
a[0]	0x10	0A	00	00	00	0B	00	00
a[2]	0x18	0C	00	00	00			
a[4]	0x20				5F	01	00	00
0x28	AD	0B	00	00				
0x30								
0x38								
p	0x40	18	00	00	00	00	00	00
0x48								
0x50								